Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	Steady Bom Bi Bi
Selected Domain	Domain 2
Selected SDG Track (1–17)	8, 9, 10
Project / Solution Name	BeliKongsi
Name (Leader)	Shawn Lee Yen Zheng
Email & Phone (Leader)	shawnlee0206@gmail.com (010-2989939)
University Affiliation(s)	Asia Pacific University, Sunway University
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Although businesses could combine their orders to reach wholesale quantities, organising group purchases creates significant recurring busywork. Owners or staff must collect quantities through WhatsApp, calculate totals, negotiate with suppliers, split deliveries and bills, collect payments, and chase late payers. Manual calculations can also lead to incorrect quantities, short deliveries, overpayments and disputes. Existing tools such as WhatsApp groups, paper records and spreadsheets do not provide a shared, reliable record of orders, payments, deliveries and price tiers.

This problem affects both micro F&B businesses and suppliers. Businesses face unnecessary purchasing costs and administrative effort, while suppliers must handle many small orders, carry credit risk and manage inefficient delivery routes.

The issue therefore has a clear economic impact: small businesses struggle to access the savings available through bulk purchasing while spending valuable time coordinating orders. BeliKongsi addresses this recurring operational burden by enabling businesses to pool orders, reach wholesale price tiers and manage the process through a shared, structured workflow.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

BeliKongsi is an AI-assisted group-buying application designed for micro and small F&B businesses in Malaysia. It allows businesses to combine their supply orders to reach suppliers' minimum order quantities and access better wholesale prices.

The application uses Generative AI (Google Gemini) to eliminate the tedious manual entry of restocking. Shop owners can simply paste raw, unstructured WhatsApp messages or old restock

texts, which often mix Malay, English, and local units like "tin", "ctn", or "papan" and the AI parses them into a structured "Usual List" draft matched to catalogue items and quantities.

A pure, rules-based system (the Kongsi Engine) handles order pooling, wholesale price tiers, cost sharing, payment balances, and automated refunds to ensure calculations remain 100% consistent, traceable, and free from AI hallucinations. Business owners retain full control by reviewing and approving the drafted orders before payment is confirmed.

The application provides a shared dashboard where participating businesses track their orders, verified wholesale savings, payment status, and delivery progress, eliminating the need to coordinate through fragmented chat groups and spreadsheets.

To develop the prototype, we apply technical skills gained from the SDC Phase 1 workshops, including integrating GenAI APIs for natural language parsing, designing a robust PostgreSQL database with Prisma ORM, and using cloud hosting to make the application accessible.

Through AI-assisted order extraction and a structured group-buying workflow, BeliKongsi aims to reduce administrative work, prevent calculation errors, and help small F&B businesses access wholesale bulk savings.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Planned Technologies

- **Frontend:** Next.js (React 19) and Tailwind CSS for a mobile-first user interface tailored for shop owners, staff, suppliers, and ops, integrating Google Maps for regional pickup hubs.
- **Backend:** Next.js App Router server endpoints with TypeScript and Zod validation, executing the deterministic Kongsi Engine for order pooling, wholesale tier calculations, and ledger management.
- **Database:** Supabase PostgreSQL (via Prisma ORM) with Row-Level Security (RLS) to store business profiles, products, group orders, price tiers, and tamper-evident audit logs.
- **AI Integration:** Google Gemini API (@google/genai) to parse messy, multi-lingual WhatsApp restock messages into structured draft orders.
- **Cloud Hosting:** Vercel serverless platform for hosting, automated deployment, and scheduled weekly cron jobs.

System Architecture

The frontend allows business owners to save a usual list via AI-assisted WhatsApp parsing and approve weekly group purchases. The backend processes requests, applies deterministic rules for order pooling and cost sharing, and communicates with the database. The Gemini AI API parses messy text into draft items for human confirmation. The database tracks order information and group progress. The deployed application connects these components into one unified workflow.

SDC Sponsor-Provided Technologies

Our team leveraged official SDC sponsor-provided **Replit** credits. Replit and Replit Agent were utilized during prototype development for cloud development environments, full-stack scaffolding, component prototyping, and presentation asset design.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applies design thinking by focusing on the needs of micro and small F&B businesses in Malaysia. We identify their challenges in group purchasing, including manual order collection, cost calculations, payment coordination and delivery tracking. These problems guide the design of BeliKongsi's features.

To ensure the solution is user-friendly, we plan to develop a simple interface that allows business owners to join group purchases, review order quantities, check estimated savings and track order progress. Human approval will be maintained for important purchasing decisions.

For technical feasibility, we will build a functional prototype using sample business, product and supplier data. We will test the order-pooling process, wholesale price calculations, cost sharing and order status updates to identify errors and improve reliability.

For Phase 2 deployment, we will refine the interface based on user feedback, improve data validation and access control, and prepare the application for cloud deployment. Integrations with external services will be tested where available, while simulated integrations will be clearly identified. This iterative approach will help us develop a practical, reliable and scalable solution for small F&B businesses.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary SDG:** SDG 8 - Decent Work and Economic Growth **Target 8.3:** Promote development-oriented policies that support productive activities, entrepreneurship and the growth of micro, small and medium-sized enterprises (MSMEs). BeliKongsi supports this target by helping micro and small F&B businesses access potential wholesale savings through group purchasing and reducing the administrative work involved in coordinating orders. **Official source:** United Nations, Sustainable Development Goal 8 https://sdgs.un.org/goals/goal8
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In ACCCIM's 2024/2025 Malaysia Business and Economic Conditions Survey, 50.0% of respondents identified high operating costs and cash-flow problems as factors negatively affecting business performance. This highlights the financial pressure faced by businesses, although the survey does not specifically measure costs caused by small purchasing quantities or manual group-buying coordination.

Metric Category	Your Response
	Source: ACCCIM, Malaysia's Business and Economic Conditions Survey (2H 2024 and 1H 2025F) https://www.acccim.my/press-releases/acccims-press-release-on-the-release-of-acccim-malaysias-business-and-economic-conditions-survey-m-becs-report-2h-2024-and-1h-2025f-2/
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage reduction in purchasing cost per unit compared with each business's usual individual purchase price for the same product and quantity. **Measurement:** Cost savings (%) = [(Usual unit price − Group-buying unit price) ÷ Usual unit price] × 100% The KPI will be measured using supplier price data before and after group purchasing. For the prototype, sample supplier prices will be used to demonstrate the calculation. Actual savings will need to be validated through real purchase data during user testing.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Demonstrate a 10% reduction in purchasing cost per unit through group buying compared with individual purchasing, using the same products, comparable quantities and quality, and including relevant delivery fees. The target will be evaluated using sample supplier pricing in the prototype and validated against actual prices if available.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	**Type (code/dataset/model)**	**License**
Google Gemini 3.5 Flash Lite / 2.5 Flash	Model	Proprietary (Google APIs Terms of Service)
FLUX.2-klein-4b *(Black Forest Labs via NVIDIA API)*	Model	Non-Commercial / NVIDIA NIM Terms of Service
Next.js	Code	MIT License
React / React DOM	Code	MIT License
TypeScript	Code	Apache 2.0
Prisma ORM	Code	Apache 2.0
Tailwind CSS	Code	MIT License
Radix UI / shadcn/ui	Code	MIT License
@google/genai SDK	Code	Apache 2.0

@supabase/supabase-js	Code	MIT License
@vis.gl/react-google-maps	Code	MIT License
Zod	Code	MIT License
Lucide Icons (lucide-react)	Code	ISC / MIT License
Sharp	Code	Apache 2.0
node-postgres (pg)	Code	MIT License
Vitest & Playwright	Code	MIT License / Apache 2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Shawn

25/10/2026